

DOCUMENTO DE PRUEBAS DE ACEPTACIÓN

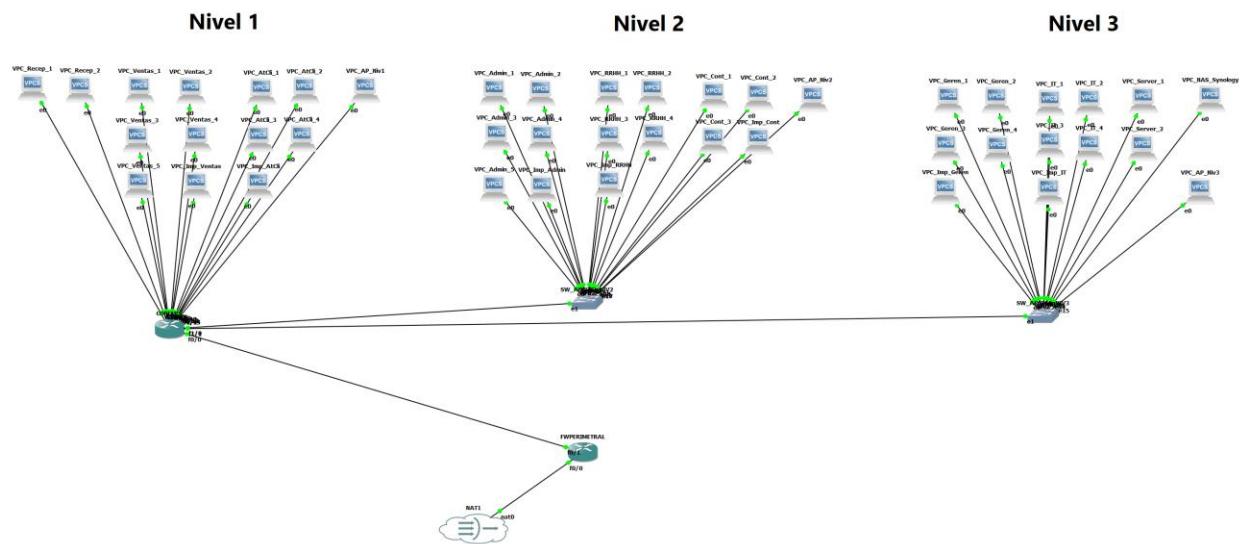
Proyecto: Infraestructura de Red – Banco ALIANZA Servicios Financieros, S.A. de C.V

1. INTRODUCCIÓN

El presente documento tiene como objetivo demostrar el correcto funcionamiento y la validación técnica de la nueva infraestructura de red diseñada para **Banco ALIANZA Servicios Financieros, S.A. de C.V.** La topología se basa en el modelo jerárquico de tres capas de Cisco (Acceso, Distribución/Core y Perímetro), integrando enrutamiento Inter-VLAN, servicios automatizados (DHCP), protocolos de enrutamiento dinámico (OSPF), políticas de seguridad (ACLs) y salida a Internet mediante traducción de direcciones (NAT).

2. TOPOLOGÍA DE RED

A continuación, se presenta el diagrama físico/lógico de la infraestructura implementada en GNS3.



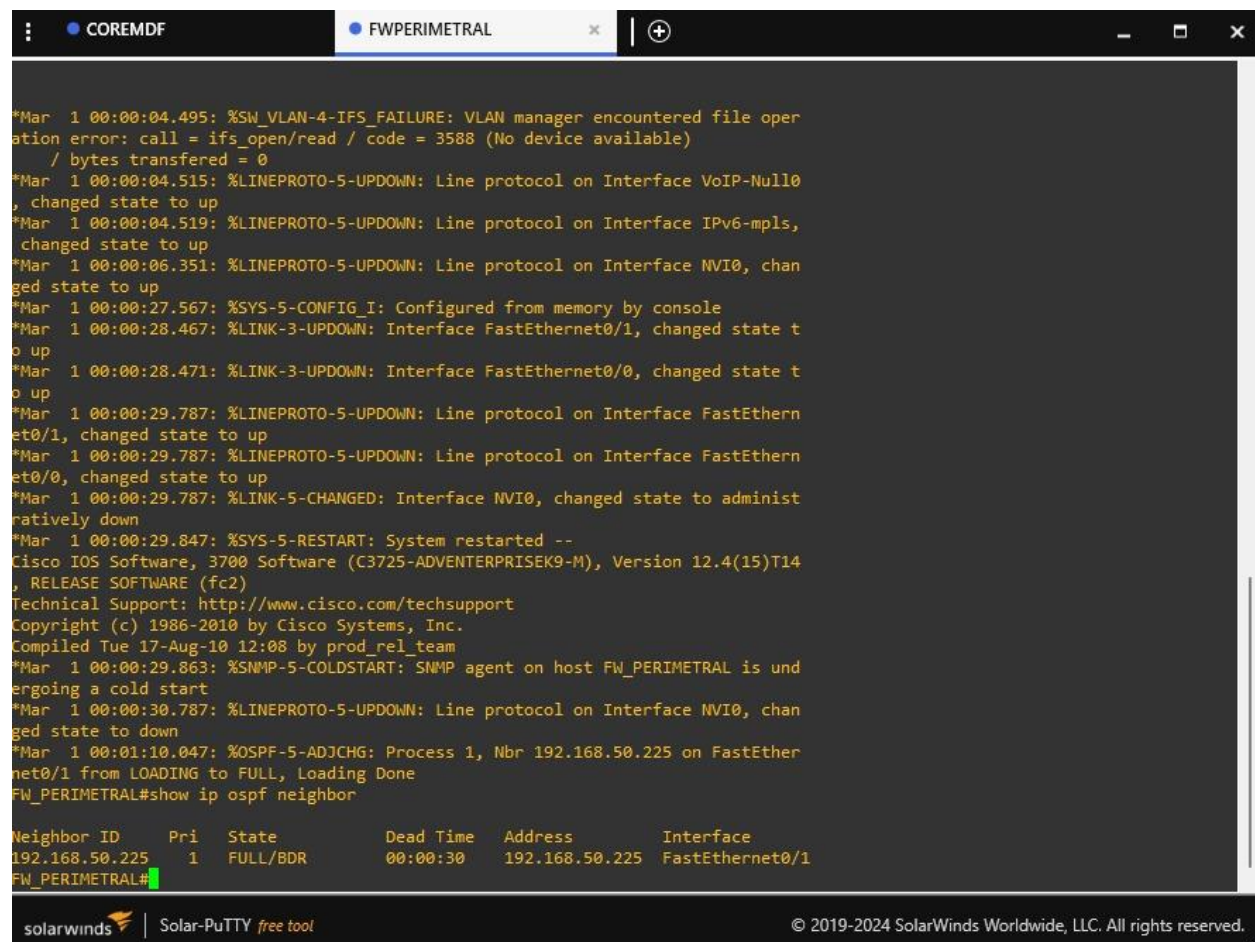
3. EVIDENCIAS DE FUNCIONAMIENTO (PRUEBAS DE ACEPTACIÓN)

3.1. Enrutamiento Dinámico (OSPF)

Objetivo: Comprobar que el Router Core y el Firewall Perimetral han establecido una adyacencia exitosa y están intercambiando rutas.

Comando ejecutado: show ip ospf neighbor (en FW_PERIMETRAL)

Resultado: Se observa el estado FULL/BDR, confirmando la sincronización de las bases de datos de ruteo.



```

COREMDF  FWPERIMETRAL
*Mar 1 00:00:04.495: %SW_VLAN-4-IFS_FAILURE: VLAN manager encountered file oper
ation error: call = ifs_open/read / code = 3588 (No device available)
/ bytes transferred = 0
*Mar 1 00:00:04.515: %LINEPROTO-5-UPDOWN: Line protocol on Interface VoIP-Null0
, changed state to up
*Mar 1 00:00:04.519: %LINEPROTO-5-UPDOWN: Line protocol on Interface IPv6-mpls,
changed state to up
*Mar 1 00:00:06.351: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, chan
ged state to up
*Mar 1 00:00:27.567: %SYS-5-CONFIG_I: Configured from memory by console
*Mar 1 00:00:28.467: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state t
o up
*Mar 1 00:00:28.471: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state t
o up
*Mar 1 00:00:29.787: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/1, changed state to up
*Mar 1 00:00:29.787: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthern
et0/0, changed state to up
*Mar 1 00:00:29.787: %LINK-5-CHANGED: Interface NVI0, changed state to administ
ratively down
*Mar 1 00:00:29.847: %SYS-5-RESTART: System restarted --
Cisco IOS Software, 3700 Software (C3725-ADVENTERPRISEK9-M), Version 12.4(15)T14
, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2010 by Cisco Systems, Inc.
Compiled Tue 17-Aug-10 12:08 by prod_rel_team
*Mar 1 00:00:29.863: %SNMP-5-COLDSTART: SNMP agent on host FW_PERIMETRAL is und
ergoing a cold start
*Mar 1 00:00:30.787: %LINEPROTO-5-UPDOWN: Line protocol on Interface NVI0, chan
ged state to down
*Mar 1 00:01:10.047: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.50.225 on FastEther
net0/1 from LOADING to FULL, Loading Done
FW_PERIMETRAL#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.50.225  1     FULL/BDR        00:00:30    192.168.50.225 FastEthernet0/1
FW_PERIMETRAL#
```

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3.2. Tabla de Enrutamiento y Subinterfaces (SVIs)

Objetivo: Validar que el Core conoce todas las subredes de las distintas áreas de la empresa (VLANs 10 a la 200) y tiene una ruta por defecto hacia Internet.

Comando ejecutado: show ip route (en CORE_MDF)

Resultado: Se listan todas las redes como conectadas directamente (C) y se visualiza la ruta por defecto inyectada por el Firewall (O*E2).

```

COREMDF
FWPERIMETRAL
Vlan80      192.168.50.129 YES NVRAM up      up
Vlan90      192.168.50.161 YES NVRAM up      up
Vlan99      192.168.50.209 YES NVRAM up      up
Vlan100     192.168.50.33  YES NVRAM up      up
Vlan200     192.168.50.193 YES NVRAM up      up
CORE_MDF#
CORE_MDF#
CORE_MDF#
CORE_MDF#
CORE_MDF#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.50.226 to network 0.0.0.0

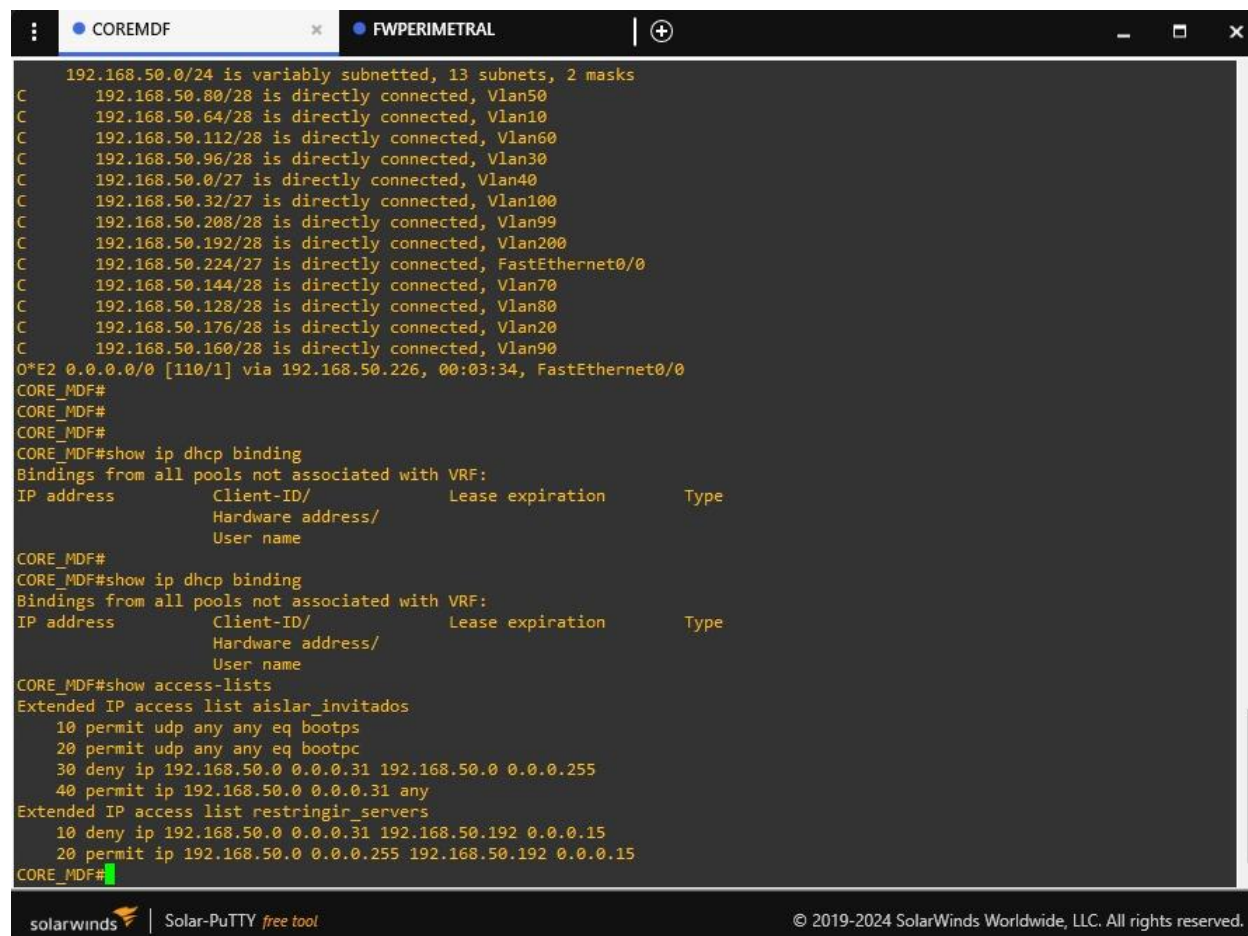
192.168.50.0/24 is variably subnetted, 13 subnets, 2 masks
C       192.168.50.80/28 is directly connected, Vlan50
C       192.168.50.64/28 is directly connected, Vlan10
C       192.168.50.112/28 is directly connected, Vlan60
C       192.168.50.96/28 is directly connected, Vlan30
C       192.168.50.0/27 is directly connected, Vlan40
C       192.168.50.32/27 is directly connected, Vlan100
C       192.168.50.208/28 is directly connected, Vlan99
C       192.168.50.192/28 is directly connected, Vlan200
C       192.168.50.224/27 is directly connected, FastEthernet0/0
C       192.168.50.144/28 is directly connected, Vlan70
C       192.168.50.128/28 is directly connected, Vlan80
C       192.168.50.176/28 is directly connected, Vlan20
C       192.168.50.160/28 is directly connected, Vlan90
O*E2 0.0.0.0/0 [110/1] via 192.168.50.226, 00:03:34, FastEthernet0/0
CORE_MDF#
CORE_MDF#
CORE_MDF#
CORE_MDF#
```

3.3. Asignación Dinámica de Direcciones (DHCP)

Objetivo: Demostrar que el servidor DHCP interno está activo y entregando concesiones IP correctamente a los dispositivos de la capa de acceso.

Comando ejecutado: show ip dhcp binding (en CORE_MDF)

Resultado: Tabla de asignaciones donde se vincula la IP entregada con la dirección MAC de los clientes.



```
192.168.50.0/24 is variably subnetted, 13 subnets, 2 masks
C 192.168.50.80/28 is directly connected, Vlan50
C 192.168.50.64/28 is directly connected, Vlan10
C 192.168.50.112/28 is directly connected, Vlan60
C 192.168.50.96/28 is directly connected, Vlan30
C 192.168.50.0/27 is directly connected, Vlan40
C 192.168.50.32/27 is directly connected, Vlan100
C 192.168.50.208/28 is directly connected, Vlan99
C 192.168.50.192/28 is directly connected, Vlan200
C 192.168.50.224/27 is directly connected, FastEthernet0/0
C 192.168.50.144/28 is directly connected, Vlan70
C 192.168.50.128/28 is directly connected, Vlan80
C 192.168.50.176/28 is directly connected, Vlan20
C 192.168.50.160/28 is directly connected, Vlan90
O*E2 0.0.0.0/0 [110/1] via 192.168.50.226, 00:03:34, FastEthernet0/0
CORE_MDF#
CORE_MDF#
CORE_MDF#
CORE_MDF#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name
CORE_MDF#
CORE_MDF#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name
CORE_MDF#show access-lists
Extended IP access list aislar_invitados
 10 permit udp any any eq bootps
 20 permit udp any any eq bootpc
 30 deny ip 192.168.50.0 0.0.0.31 192.168.50.0 0.0.0.255
 40 permit ip 192.168.50.0 0.0.0.31 any
Extended IP access list restringir_servers
 10 deny ip 192.168.50.0 0.0.0.31 192.168.50.192 0.0.0.15
 20 permit ip 192.168.50.0 0.0.0.255 192.168.50.192 0.0.0.15
CORE_MDF#
```

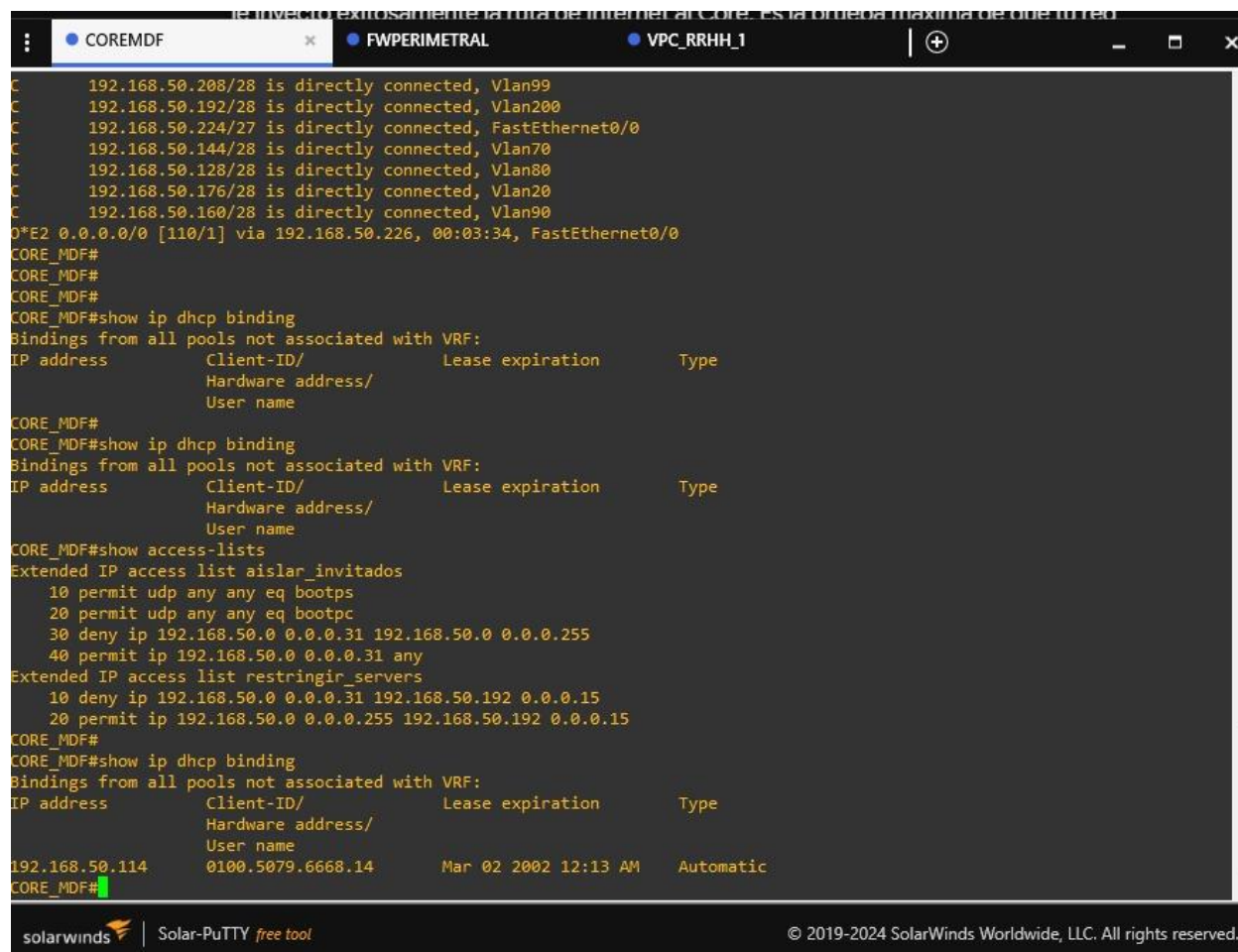
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3.4. Auditoría de Seguridad (Listas de Control de Acceso)

Objetivo: Evidenciar que las reglas de seguridad perimetral están bloqueando el tráfico no autorizado (ej. Invitados intentando acceder a la red corporativa o servidores).

Comando ejecutado: show access-lists (en CORE_MDF)

Resultado: Se visualizan las reglas configuradas y los contadores de coincidencias (matches), demostrando que el router interceptó e impidió intentos de conexión no permitidos.



```

C 192.168.50.208/28 is directly connected, Vlan99
C 192.168.50.192/28 is directly connected, Vlan200
C 192.168.50.224/27 is directly connected, FastEthernet0/0
C 192.168.50.144/28 is directly connected, Vlan70
C 192.168.50.128/28 is directly connected, Vlan80
C 192.168.50.176/28 is directly connected, Vlan20
C 192.168.50.160/28 is directly connected, Vlan90
O*E2 0.0.0.0/0 [110/1] via 192.168.50.226, 00:03:34, FastEthernet0/0
CORE_MDF#
CORE_MDF#
CORE_MDF#
CORE_MDF#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name

CORE_MDF#
CORE_MDF#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name

CORE_MDF#show access-lists
Extended IP access list aislar_invitados
 10 permit udp any any eq bootps
 20 permit udp any any eq bootpc
 30 deny ip 192.168.50.0 0.0.0.31 192.168.50.0 0.0.0.255
 40 permit ip 192.168.50.0 0.0.0.31 any
Extended IP access list restringir_servers
 10 deny ip 192.168.50.0 0.0.0.31 192.168.50.192 0.0.0.15
 20 permit ip 192.168.50.0 0.0.0.255 192.168.50.192 0.0.0.15
CORE_MDF#
CORE_MDF#show ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration    Type
                Hardware address/
                User name
192.168.50.114   0100.5079.6668.14  Mar 02 2002 12:13 AM  Automatic
CORE_MDF#
```

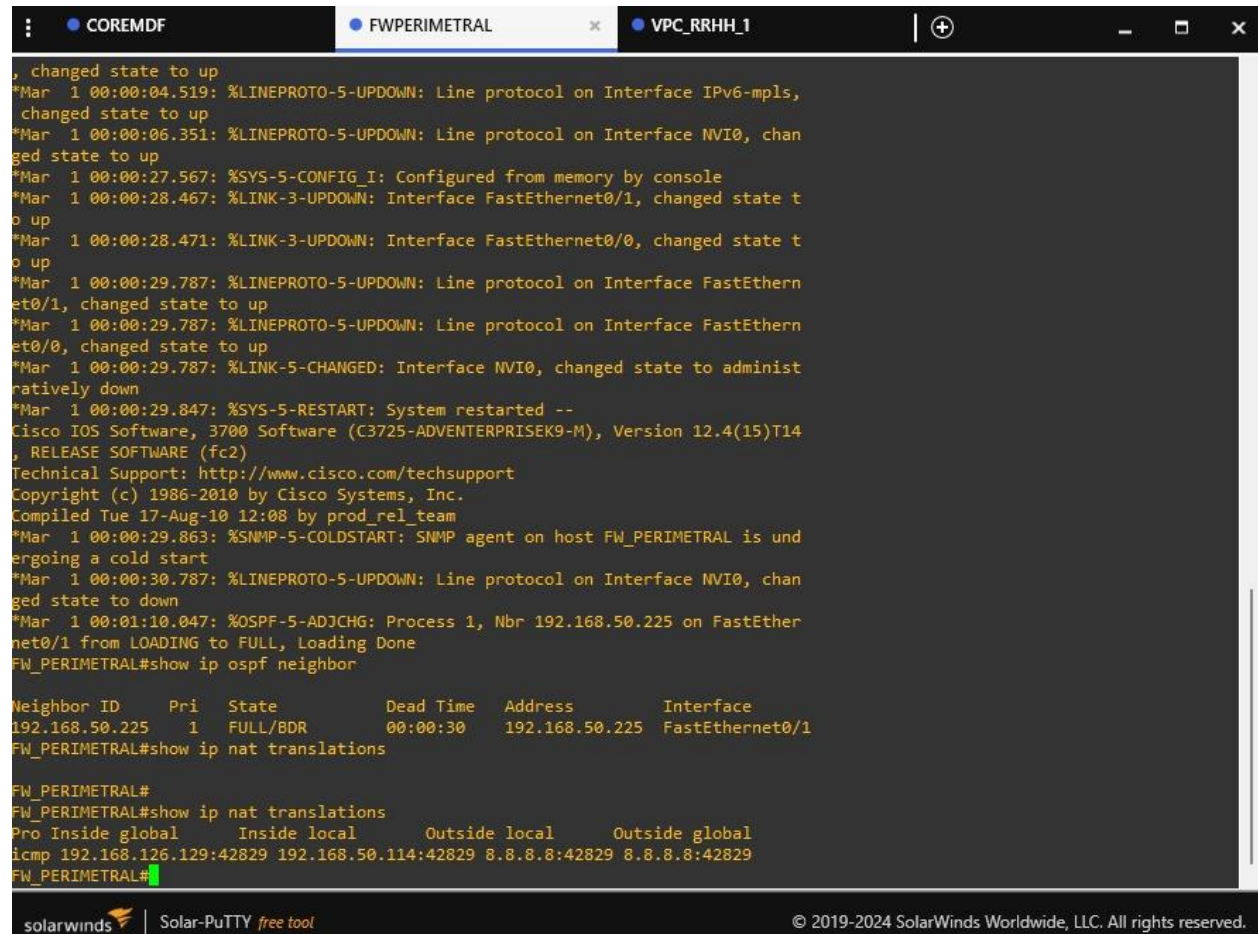
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3.5. Traducción de Direcciones de Red (NAT / PAT)

Objetivo: Validar que el Firewall enmascara exitosamente las IPs privadas de la empresa utilizando su IP pública asignada para permitir la navegación hacia el exterior.

Comando ejecutado: show ip nat translations (en FW_PERIMETRAL)

Resultado: Se observa la tabla de traducción activa durante un intento de conexión hacia una IP pública.



The screenshot shows a SolarWinds Solar-PuTTY terminal window with three tabs: COREMDF, FWPERIMETRAL (active), and VPC_RRHH_1. The terminal displays a series of Cisco IOS log messages, including interface status changes and system restarts. The user has entered the command 'show ip nat translations' at the FW_PERIMETRAL prompt, resulting in the following output:

```
FW_PERIMETRAL#show ip nat translations
Pro Inside global      Inside local      Outside local      Outside global
icmp 192.168.126.129:42829 192.168.50.114:42829 8.8.8.8:42829 8.8.8.8:42829
FW_PERIMETRAL#
```

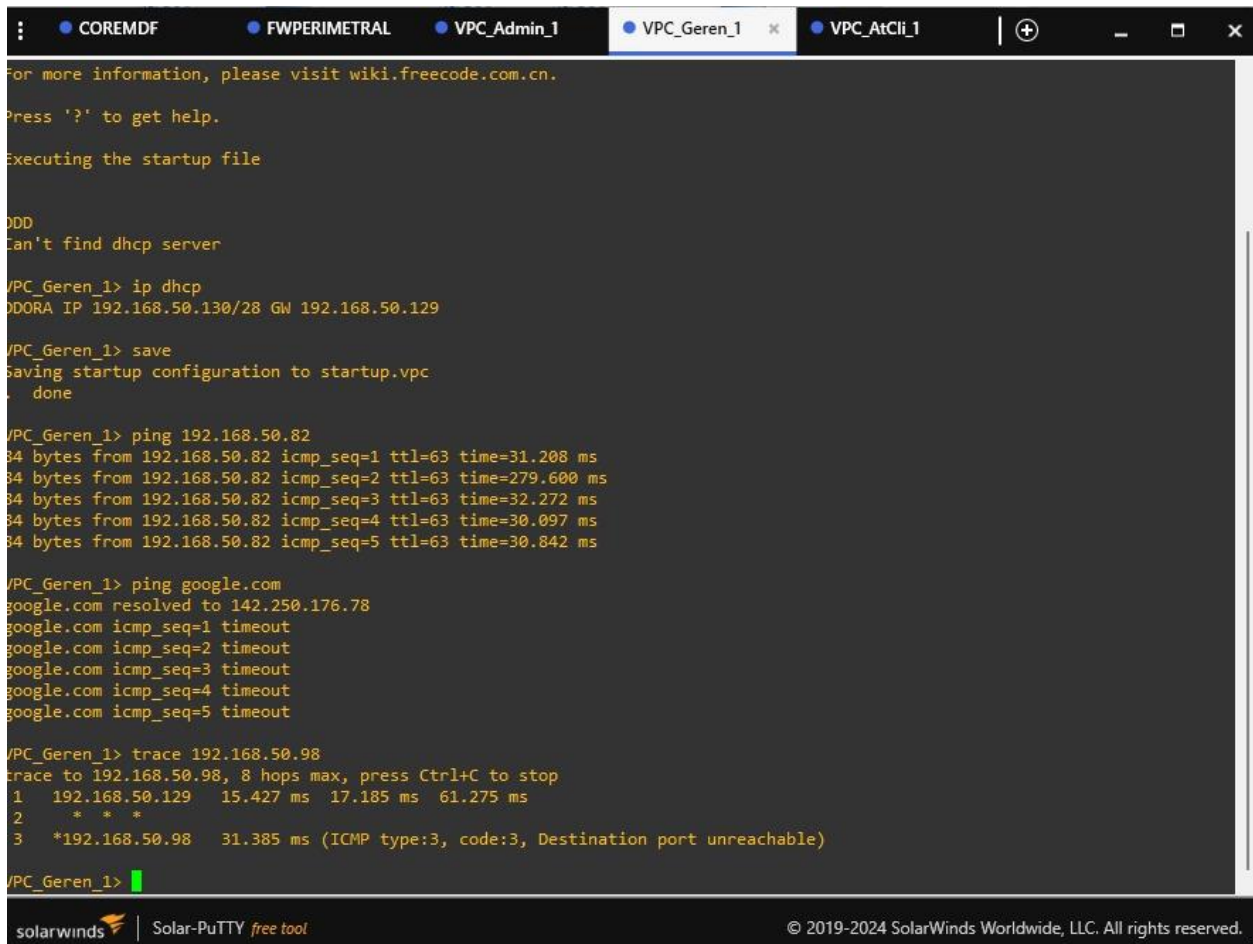
The bottom of the window shows the SolarWinds logo and the text 'Solar-PuTTY free tool' on the left, and the copyright notice '© 2019-2024 SolarWinds Worldwide, LLC. All rights reserved.' on the right.

3.6. Conectividad Extremo a Extremo (Pruebas de Usuario)

Objetivo: Comprobar la experiencia del usuario final realizando consultas a servicios externos (Internet) y servicios internos (Servidores).

Comandos ejecutados: ping hacia admin, ping hacia cliente y ping google.com / trace google.com (en PC de Gerencia).

Resultado: Respuesta exitosa desde la sala de servidores y resolución de nombres de dominio (DNS) exitosa hacia Internet, pasando por todos los saltos de red internos.



```
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

DDD
Can't find dhcp server

/PC_Geren_1> ip dhcp
DDORA IP 192.168.50.130/28 GW 192.168.50.129

/PC_Geren_1> save
Saving startup configuration to startup.vpc
. done

/PC_Geren_1> ping 192.168.50.82
34 bytes from 192.168.50.82 icmp_seq=1 ttl=63 time=31.208 ms
34 bytes from 192.168.50.82 icmp_seq=2 ttl=63 time=279.600 ms
34 bytes from 192.168.50.82 icmp_seq=3 ttl=63 time=32.272 ms
34 bytes from 192.168.50.82 icmp_seq=4 ttl=63 time=30.097 ms
34 bytes from 192.168.50.82 icmp_seq=5 ttl=63 time=30.842 ms

/PC_Geren_1> ping google.com
google.com resolved to 142.250.176.78
google.com icmp_seq=1 timeout
google.com icmp_seq=2 timeout
google.com icmp_seq=3 timeout
google.com icmp_seq=4 timeout
google.com icmp_seq=5 timeout

/PC_Geren_1> trace 192.168.50.98
Trace to 192.168.50.98, 8 hops max, press Ctrl+C to stop
 1  192.168.50.129    15.427 ms  17.185 ms  61.275 ms
 2      * * *
 3  *192.168.50.98   31.385 ms (ICMP type:3, code:3, Destination port unreachable)

/PC_Geren_1> 
```

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4. CONCLUSIÓN

Tras ejecutar el plan de pruebas, se dictamina que la red de **Banco ALIANZA Servicios Financieros, S.A. de C.V** es 100% operativa. Cumple con todos los requerimientos de segmentación (VLANs), alta disponibilidad lógica, seguridad y salida a Internet establecidos en el diseño original.